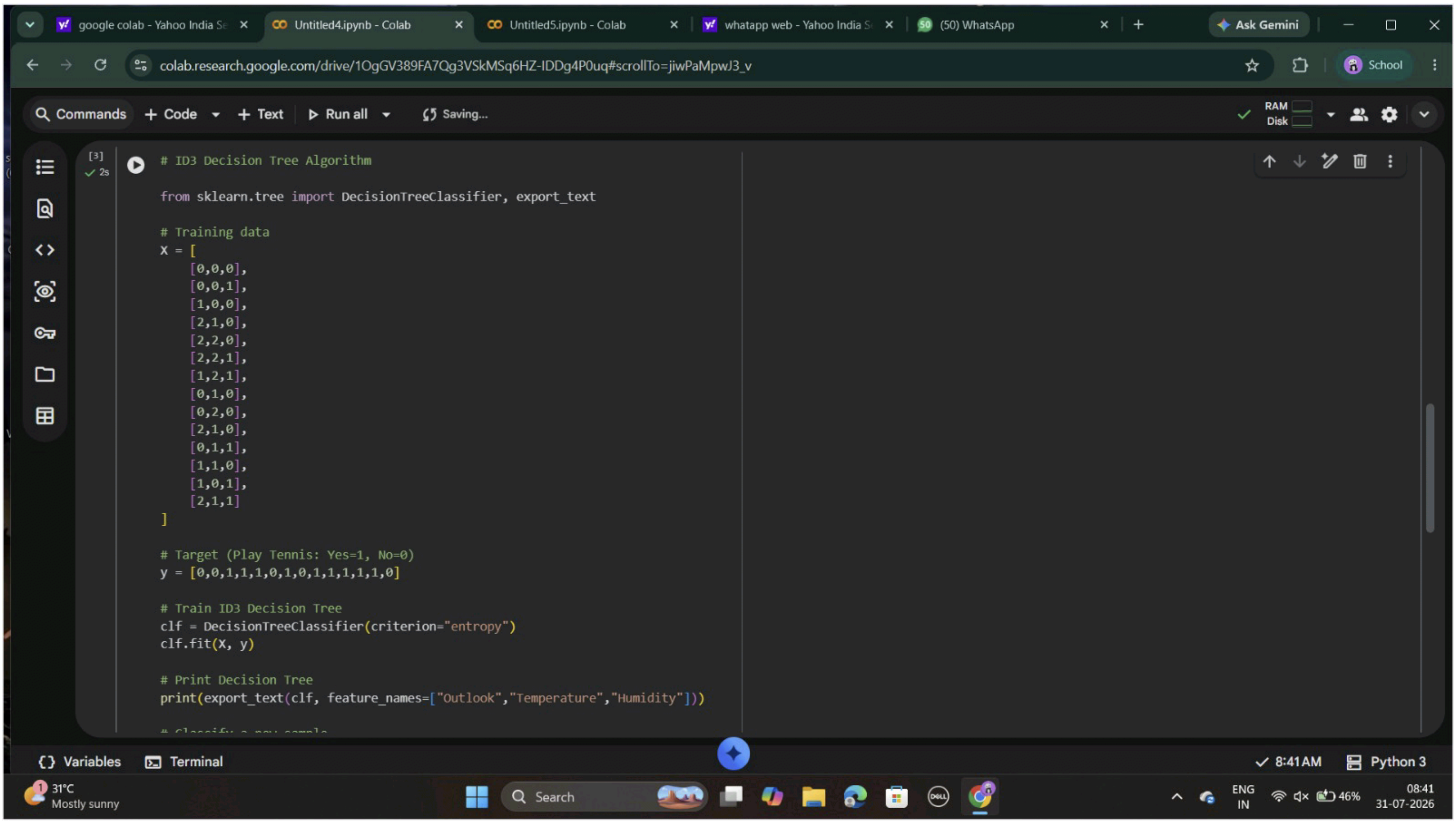

```python
# ID3 Decision Tree Algorithm

from sklearn.tree import DecisionTreeClassifier, export_text

# Training data
X = [
    [0,0,0],
    [0,0,1],
    [1,0,0],
    [2,1,0],
    [2,2,0],
    [2,2,1],
    [1,2,1],
    [0,1,0],
    [0,2,0],
    [2,1,0],
    [0,1,1],
    [1,1,0],
    [1,0,1],
    [2,1,1]
]

# Target (Play Tennis: Yes=1, No=0)
y = [0,0,1,1,1,0,1,0,1,1,1,1,1,0]

# Train ID3 Decision Tree
clf = DecisionTreeClassifier(criterion="entropy")
clf.fit(X, y)

# Print Decision Tree
print(export_text(clf, feature_names=["Outlook","Temperature","Humidity"]))
```

```python
clf = DecisionTreeClassifier(criterion="entropy")
clf.fit(X, y)

# Print Decision Tree
print(export_text(clf, feature_names=["Outlook","Temperature","Humidity"]))

# Classify a new sample
new_sample = [[0,1,0]]
prediction = clf.predict(new_sample)

print("Prediction:", "Yes" if prediction[0] == 1 else "No")
```

```
|--- Outlook <= 0.50
|   |--- Temperature <= 0.50
|   |   |--- class: 0
|   |--- Temperature >  0.50
|   |   |--- Humidity <= 0.50
|   |   |   |--- Temperature <= 1.50
|   |   |   |   |--- class: 0
|   |   |   |--- Temperature >  1.50
|   |   |   |   |--- class: 1
|   |   |--- Humidity >  0.50
|   |   |   |--- class: 1
|--- Outlook >  0.50
|   |--- Humidity <= 0.50
|   |   |--- class: 1
|   |--- Humidity >  0.50
|   |   |--- Outlook <= 1.50
|   |   |   |--- class: 1
|   |   |--- Outlook >  1.50
|   |   |   |--- class: 0

Prediction: No
```